MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 01 (SOLUTIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. In an AP, if $d = -4$, $n = 7$, and $a_n = 4$, then the first term a is: [1]

(A) 6 (B) 7 (C) 20 (D) 28

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(D) 28** [0.5 Mark]

Given: Common difference $d = -4$, number of terms $n = 7$, and n^{th} term $a_n = 4$.
We know the formula for the nth term of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]
Substituting the given values step by step:

$$4 = a + (7-1)(-4) \implies 4 = a + 6(-4) \implies 4 = a - 24$$

Transposing -24 to the LHS:

$$a = 4 + 24 = \mathbf{28}$$

Q2. In an AP, if $a = 3.5$, $d = 0$, and $n = 101$, then a_n equals: [1]

(A) 0 (B) 3.5 (C) 103.5 (D) 104.5

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 3.5** [0.5 Mark]

Given: First term $a = 3.5$, common difference $d = 0$, term number $n = 101$.
We know the general term formula of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]
Substituting $a = 3.5$, $n = 101$, and $d = 0$:

$$a_{101} = 3.5 + (101-1) \times 0 = 3.5 + 100 \times 0 = 3.5 + 0 = \mathbf{3.5}$$

Q3. The list of numbers $-10, -6, -2, 2, \ldots$ is: [1]

(A) an AP with $d = -16$ (B) an AP with $d = 4$ (C) an AP with $d = -4$ (D) not an AP

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) an AP with d = 4** [0.5 Mark]

We check whether consecutive differences are constant:

Formula / Theorem Used: A sequence is an AP if $d = a_{k+1} - a_k$ is identical for all consecutive pairs.

Justification: [0.5 Mark]

Here $a_1 = -10$, $a_2 = -6$, $a_3 = -2$, $a_4 = 2$:

$$a_2 - a_1 = -6 - (-10) = -6 + 10 = 4, \quad a_3 - a_2 = -2 - (-6) = 4, \quad a_4 - a_3 = 2 - (-2) = 4$$

Since the difference is constant ($d = 4$), it forms an AP with $d = 4$.

Q4. The 11th term of the AP $-3, -\frac{1}{2}, 2, \ldots$ is: [1]

(A) 28 (B) 22 (C) -38 (D) $-46\frac{1}{2}$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 22** [0.5 Mark]

Given: First term $a = -3$, second term $a_2 = -\frac{1}{2}$, so $d = -\frac{1}{2} - (-3) = -\frac{1}{2} + \frac{6}{2} = \frac{5}{2}$.

We know the general term formula of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

Substituting $a = -3$, $n = 11$, and $d = \frac{5}{2}$:

$$a_{11} = -3 + (11 - 1)\left(\frac{5}{2}\right) = -3 + 10 \times \frac{5}{2} = -3 + 5 \times 5 = -3 + 25 = \mathbf{22}$$

Q5. **Assertion (A):** The sequence $-10, -6, -2, 2, \ldots$ forms an AP with common difference 4.
Reason (R): A sequence $a_1, a_2, a_3, \ldots$ is an AP if $a_{k+1} - a_k$ is constant for all $k \geq 1$. [1]

(A) Both (A) and (R) are true; (R) explains (A). (B) Both are true; (R) does not explain (A).
(B) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true; (R) explains (A).** [0.5 Mark]

We verify the definition of an arithmetic progression:

Formula / Theorem Used: A list of numbers is an AP iff $a_{k+1} - a_k = d$ (a fixed constant) for every $k \geq 1$.

Justification: [0.5 Mark]

Testing consecutive differences: $-6 - (-10) = 4$, $-2 - (-6) = 4$, $2 - (-2) = 4$. Thus (A) is true, (R) states the exact defining criterion of an AP (true), and (R) correctly explains (A).

Q6. Find the 20th term of the AP $4, 9, 14, 19, \ldots$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying a and d [1 Mark]

Given: AP is $4, 9, 14, 19, \ldots$. Here, first term $a = 4$, common difference $d = 9 - 4 = 5$, and $n = 20$.
We know the formula for the nth term of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 2: Substituting Values to Find a_{20} [1 Mark]

$$a_{20} = 4 + (20 - 1) \times 5 = 4 + 19 \times 5 = 4 + 95 = \boxed{\mathbf{99}}$$

Q7. The sum of the 4th and 8th terms of an AP is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the AP. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let first term $= a$ and common difference $= d$. Given $a_4 + a_8 = 24$ and $a_6 + a_{10} = 44$.
We know the general term formula of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Forming Two Linear Equations in a and d [1.5 Marks]

From $a_4 + a_8 = 24$:

$$[a + 3d] + [a + 7d] = 24 \implies 2a + 10d = 24 \implies a + 5d = 12 \quad \text{— (i)}$$

From $a_6 + a_{10} = 44$:

$$[a + 5d] + [a + 9d] = 44 \implies 2a + 14d = 44 \implies a + 7d = 22 \quad \text{— (ii)}$$

Step 2: Solving Equations (i) and (ii) for d and a [1 Mark]

Subtracting (i) from (ii):

$$(a + 7d) - (a + 5d) = 22 - 12 \implies 2d = 10 \implies d = \frac{10}{2} = 5$$

Substituting $d = 5$ into (i):

$$a + 5(5) = 12 \implies a + 25 = 12 \implies a = 12 - 25 = -13$$

Step 3: Writing the First Three Terms [0.5 Mark]

$a_1 = a = -13, \quad a_2 = a + d = -13 + 5 = -8, \quad a_3 = a + 2d = -13 + 10 = -3.$

First three terms of the AP: $\mathbf{-13}$, $\mathbf{-8}$, $\mathbf{-3}$

Q8. The pth term of an AP is q and its qth term is p. Prove that its nth term is $(p+q-n)$, and deduce that its $(p+q)^{\text{th}}$ term is zero. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: In an AP with first term a and common difference d ($p \neq q$): $a_p = q$ and $a_q = p$.
We know the general term formula of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Finding Common Difference d and First Term a [1.5 Marks]

$$a_p = a + (p-1)d = q \quad \text{— (i)}$$
$$a_q = a + (q-1)d = p \quad \text{— (ii)}$$

Subtracting (ii) from (i):

$$[(p-1)-(q-1)]d = q - p \implies (p-q)d = -(p-q)$$

Dividing by $(p-q) \neq 0$: $d = -1$.
Substituting $d = -1$ into (i):

$$a + (p-1)(-1) = q \implies a - p + 1 = q \implies a = p + q - 1$$

Step 2: Deriving a_n and Deducing $a_{p+q} = 0$ [1.5 Marks]

$$a_n = a + (n-1)d = (p+q-1) + (n-1)(-1) = p + q - 1 - n + 1 = \boxed{p+q-n}$$

Setting $n = p + q$ in this result:

$$a_{p+q} = p + q - (p+q) = \boxed{0 \quad \text{[Hence Proved]}}$$

Q9. If the ratio of the sums of first n terms of two APs is $(7n+1) : (4n+27)$, find the ratio of their 11th terms. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: Let the first AP have first term a, common difference d, sum S_n; and the second AP have first term A, common difference D, sum S'_n. Given $\dfrac{S_n}{S'_n} = \dfrac{7n+1}{4n+27}$.
We know the formulas for sum of n terms and m^{th} term of an AP:

Formula / Theorem Used: $S_n = \dfrac{n}{2}[2a + (n-1)d]$ and $a_m = a + (m-1)d$

Step 1: Simplifying the Ratio of Sums [1 Mark]

$$\frac{S_n}{S'_n} = \frac{\frac{n}{2}[2a+(n-1)d]}{\frac{n}{2}[2A+(n-1)D]} = \frac{2a+(n-1)d}{2A+(n-1)D} = \frac{a + \left(\frac{n-1}{2}\right)d}{A + \left(\frac{n-1}{2}\right)D} = \frac{7n+1}{4n+27}$$

Step 2: Matching Coefficients for the 11th Term Ratio [2 Marks]

The ratio of 11th terms is $\dfrac{a_{11}}{A_{11}} = \dfrac{a+10d}{A+10D}$.
Setting $\dfrac{n-1}{2} = 10 \implies n - 1 = 20 \implies n = 21$ in Step 1:

$$\frac{a_{11}}{A_{11}} = \frac{a+10d}{A+10D} = \frac{7(21)+1}{4(21)+27} = \frac{147+1}{84+27} = \frac{148}{111}$$

Dividing numerator and denominator by their HCF 37 ($148 = 4 \times 37$, $111 = 3 \times 37$):

$$\boxed{\frac{a_{11}}{A_{11}} = \mathbf{4:3}}$$

Q10. CASE STUDY: Manufacturing Production Schedule (Smart TV Factory)

A consumer electronics manufacturing plant in Greater Noida produces high-definition smart LED television sets. To meet rising market demand, management modernizes the plant with automated robotic assembly lines. Consequently, the annual production of TV sets increases uniformly by a constant number of units every year. The factory produced 16,000 sets in its 6th year of commercial operation and 22,600 sets in its 9th year. [4]

(i) Find the fixed annual increase in production (d). [1]

(ii) Determine the production of smart TV sets during the first year of operation (a). [1]

(iii) Calculate the total cumulative production of smart TV sets manufactured in the first 8 years. [2]

— **OR** —

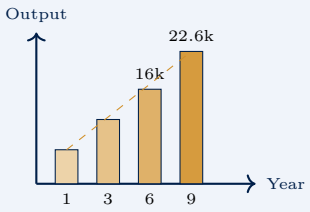
In which operational year will the annual production reach 29,200 smart TV sets? [2]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Extracted Scenario Data: Annual production forms an AP with first-year output a and uniform annual increase d. Given $a_6 = 16{,}000$ and $a_9 = 22{,}600$.

We know the general term and sum formulas of an AP:

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$



Sub-part (i): Fixed Annual Increase in Production (d) [1 Mark]

Using $a_6 = a + 5d = 16{,}000$ (eq. 1) and $a_9 = a + 8d = 22{,}600$ (eq. 2), subtract eq. 1 from eq. 2:

$$(a + 8d) - (a + 5d) = 22{,}600 - 16{,}000 \implies 3d = 6{,}600 \implies d = \frac{6{,}600}{3} = \boxed{\textbf{2,200 TV sets}}$$

Sub-part (ii): Production in the First Year (a) [1 Mark]

Substituting $d = 2{,}200$ into $a + 5d = 16{,}000$:

$$a + 5(2{,}200) = 16{,}000 \implies a + 11{,}000 = 16{,}000 \implies a = 16{,}000 - 11{,}000 = \boxed{\textbf{5,000 TV sets}}$$

Sub-part (iii): Total Cumulative Production in First 8 Years [2 Marks]

$$S_8 = \frac{8}{2}[2(5{,}000) + (8-1)(2{,}200)] = 4[10{,}000 + 7 \times 2{,}200]$$
$$= 4(10{,}000 + 15{,}400) = 4 \times 25{,}400 = \boxed{\textbf{1,01,600 TV sets}}$$

Sub-part (iii) [OR Option]: Year When Output Reaches 29,200 TV Sets [2 Marks]

Let $a_n = 29{,}200$. Substituting $a = 5{,}000$ and $d = 2{,}200$:

$$5{,}000 + (n-1)(2{,}200) = 29{,}200 \implies (n-1)(2{,}200) = 29{,}200 - 5{,}000 = 24{,}200$$

$$n - 1 = \frac{24{,}200}{2{,}200} = 11 \implies n = 11 + 1 = \boxed{\textbf{12}^{\text{th}}\textbf{ operational year}}$$

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